Fold-Angle Distance Geometry:

A Unified Framework for Nonlocal Correlation, Spatial Projection, and 3.5-Dimensional Fold States

Abstract:

We propose a geometric model in which physical distance is not the metric separation in three-dimensional space

but the angular difference between two projections of a higher-dimensional fold structure. Within this framework,

quantum nonlocality, correlated state transitions, and apparent superluminal effects emerge naturally as dual-face

projections of a single fold node.

A minimal distance–fold relation is derived:

$D(\theta) = D_{max} \cdot \sin(\theta / 2)$

where θ is the fold angle in a 3.5-dimensional legal-fold layer (LFL). We show that the familiar 3D world is a restricted

projection of this fold-layer, and that "instantaneous" quantum correlations arise when $\theta \to 0$, yielding $D \to 0$ in the actual

geometry.

Keywords:

Fold-Angle Geometry, 3.5D Legal Fold Layer, Dual-Face Entanglement, Projection Space, Nonlocal Correlation,

Distance Redefinition

1. Introduction

Modern physics faces three deep puzzles:

1) The origin of nonlocal correlations (entanglement).

2) The limitation of spatial distance as a 3D metric.

3) The apparent existence of transitions that do not obey classical spatial separation.

We introduce a framework where:

• 3D distance is not fundamental

• Quantum entanglement is a projection artifact

• Space emerges from a higher fold structure

2. Fold-Angle Distance Model

$D(\theta) = D_{max} \sin(\theta/2)$

If $\theta = 0 \rightarrow D = 0$ (dual-face entanglement)

If $\theta = \pi \rightarrow D = D_{max}$ (maximum separation)

3. The 3.5-Dimensional Legal Fold Layer (LFL)

A transitional dimensional layer with:

• minimal energy configuration

• supports fold transformations

• explains nonlocal effects without propagation

4. Dual-Face Interpretation of Correlated States

Entangled particles correspond to two projected faces of a single fold node.

5. Projection and Apparent Superluminal Effects

Distance changes arise from $\Delta\theta$ in LFL, not propagation in 3D.

6. Discussion

This model unifies several unexplained quantum phenomena using a simple geometric principle.

7. Conclusion

Distance is redefined as a fold-angle measure. Nonlocality becomes projection duality.

3.5D LFL emerges as the hidden layer of spatial organization.